SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

AI-Based Online Banking Management System

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1. Introduction

The AI-Based Online Banking Management System is a secure web application that enables customers to perform banking operations such as account management, money transfer, transaction history, loan applications, and profile management. The system also provides administrative functionalities for monitoring customer activities, detecting fraudulent transactions using Artificial Intelligence, and generating audit reports.

The objective of the system is to improve banking efficiency, enhance security, and provide intelligent decision support through AI-based analysis.

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2. Functional Requirements

REQ-001

The customer shall register using a valid email address and mobile number.

REQ-002

The system shall verify customer registration using a One-Time Password (OTP).

REQ-003

The customer shall log in using a username and password.

REQ-004

The customer shall reset forgotten passwords using OTP verification.

REQ-005

The system shall authenticate users before granting access to banking services.

REQ-006

The customer shall view account balance after successful authentication.

REQ-007

The customer shall transfer funds to registered beneficiaries.

REQ-008

The system shall validate beneficiary account details before fund transfer.

REQ-009

The system shall update account balances immediately after successful transactions.

REQ-010

The system shall generate digital transaction receipts.

REQ-011

The customer shall download monthly account statements.

REQ-012

The customer shall view complete transaction history.

REQ-013

The customer shall update personal profile information.

REQ-014

The administrator shall create customer accounts.

REQ-015

The administrator shall modify customer account information.

REQ-016

The administrator shall freeze suspicious customer accounts.

REQ-017

The administrator shall approve transactions exceeding ₹5,00,000.

REQ-018

The system shall send SMS notifications after every successful transaction.

REQ-019

The system shall send email notifications after every successful transaction.

REQ-020

The system shall detect fraudulent transactions using Artificial Intelligence.

REQ-021

The fraud detection module shall notify administrators immediately after detecting suspicious transactions.

REQ-022

The system shall maintain audit logs for every banking operation.

REQ-023

The administrator shall generate monthly audit reports.

REQ-024

The system shall encrypt all customer passwords using AES-256 encryption.

REQ-025

The system shall automatically log out inactive users after five minutes.

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3. Non-Functional Requirements

REQ-026

The system response time shall not exceed two seconds.

REQ-027

The application shall support at least 10,000 concurrent users.

REQ-028

System availability shall be at least 99.9%.

REQ-029

Daily database backup shall be performed automatically.

REQ-030

All sensitive customer information shall be encrypted during storage and transmission.

REQ-031

The application shall support modern web browsers.

REQ-032

The application shall recover automatically after unexpected failures.

REQ-033

The AI fraud detection model shall produce at least 95% prediction accuracy.

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4. Business Rules

1. Customers cannot transfer money without successful authentication.
2. Transactions above ₹5,00,000 require administrator approval.
3. Every successful transaction must generate an audit record.
4. Passwords shall never be stored in plain text.
5. Every account must be linked to one verified mobile number.
6. Fraud detection must be executed before transaction completion.

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5. Requirement Changes (Version 2.0)

The following requirement modifications are introduced in Version 2.0.

Change 1

Biometric authentication shall be introduced in addition to password authentication.

Change 2

Password-only login shall become optional after biometric enrollment.

Change 3

The administrator approval threshold shall change from ₹5,00,000 to ₹3,00,000.

Change 4

The fraud detection module shall use Deep Learning instead of the existing Machine Learning model.

Change 5

SMS notifications shall become optional.

Change 6

Email notifications shall become mandatory.

Change 7

Customers shall be allowed to schedule future fund transfers.

Change 8

The system shall recommend suspicious transaction reasons using Explainable AI.

Change 9

The audit reporting module shall include AI-generated risk scores.

Change 10

The dependency between authentication and transaction modules shall be strengthened using adaptive dependency analysis.

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6. Assumptions

- All users have valid bank accounts.
- Internet connectivity is available.
- OTP services are operational.
- AI prediction models are trained before deployment.

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7. Constraints

- The application shall comply with banking security regulations.
- Customer information shall remain confidential.
- Unauthorized access shall be prevented.
- AI recommendations shall be explainable to administrators.

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8. Expected Dependencies

Authentication depends on User Registration.

OTP Verification depends on Registration.

Login depends on OTP Verification.

Account Balance depends on Login.

Money Transfer depends on Login.

Money Transfer depends on Beneficiary Validation.

Transaction Receipt depends on Money Transfer.

Fraud Detection depends on Transaction Processing.

Audit Report depends on Audit Logs.

Notification Module depends on Transaction Completion.

Administrator Approval depends on Transaction Amount.

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9. Conflict Scenarios

Changing the authentication module affects Login, Balance Inquiry, Money Transfer, Transaction History, and Profile Management.

Changing the fraud detection algorithm affects AI prediction, Administrator Notification, Audit Reports, and Risk Assessment.

Changing the transaction approval threshold affects Transaction Processing, Administrator Approval, and Audit Logs.

Changing notification policies affects SMS Service, Email Service, and Transaction Receipt Generation.
